A Brief Intro to NN Verification

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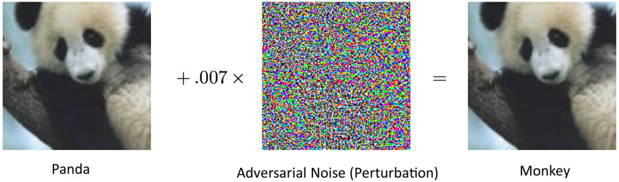
Vulnerability of Neural Networks

- Neural networks have become a crucial element in modern artificial intelligence.
- However, they are often black-boxes and can behave unexpectedly and produce surprisingly wrong results, such as **adversarial examples**:

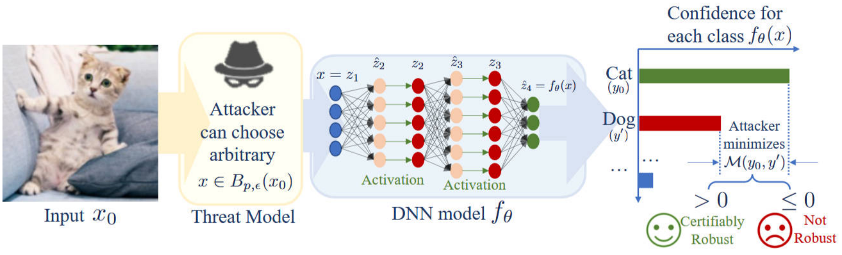


Vulnerability of Neural Networks

- In this case, we introduce the neural network verification problem, which aims to formally guarantee **properties** of **neural networks** such as **robustness**, safety, and correctness.
- **Robustness** means a neural network's ability to resist small, intentional manipulations of its input designed to trick this neural network.

Vulnerability of Neural Networks

- As the diagram shows, an attacker can create a slightly altered version of an input image (like the cat).
- A model is considered robust for that input if its confidence in the correct class ("Cat") remains higher than any incorrect class ("Dog").
- The verification process provides a guarantee that the neural network's prediction will not change within the specified threat boundaries, ensuring reliable and secure performance.



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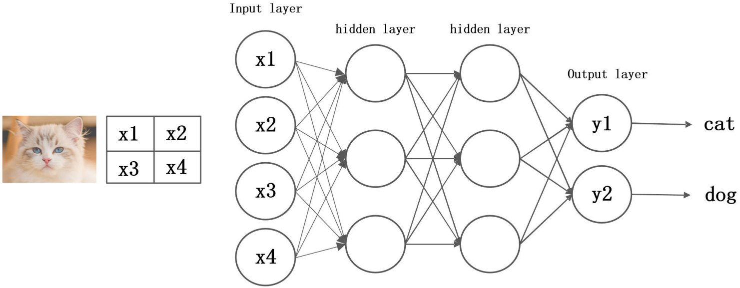
Neural Network Verification Workflow

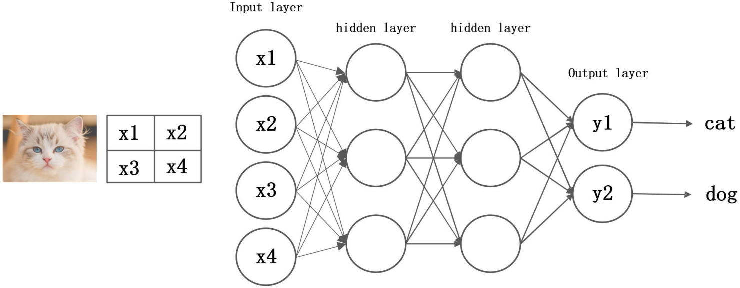
To help explain the Workflow of Neural Network Verification, let's consider a highly simplified scenario:

- Suppose the image of a cat is represented by only four pixel values—x_1, x_2, x_3, and x_4. These four pixel values serve as the input to the neural network.
- For the neural network to correctly classify the image as a cat, the output value y_1 (representing "cat") should be greater than y_2 (representing "dog").



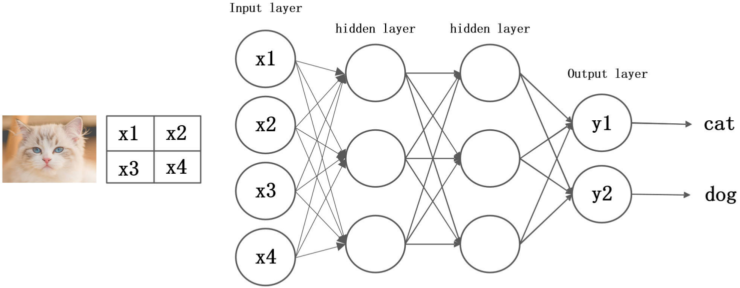
Neural Network Verification Workflow

- Assume that: $x_1 = 0.3, x_2 = 0.2, x_3 = 0.7, x_4 = 0.5$
- An attacker might subtly alter these pixel values in an attempt to fool the neural network into misclassifying the cat as a dog.
- Even with just four pixels, there are countless possible perturbations—some of which may result in the image being misclassified.



Neural Network Verification Workflow

- To assess the robustness of this neural network, we turn to neural network verification.
- For example, we may want to be absolutely certain whether modifying each pixel within a range of ± 0.02 could cause misclassification.
- Exhaustively testing all possible pixels within this range is computationally infeasible.

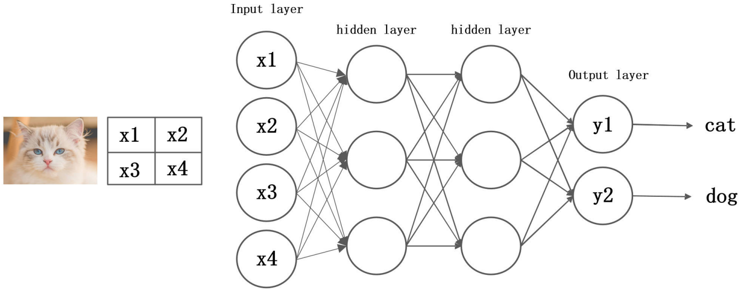


Neural Network Verification Workflow

- Instead, we set each pixel as an interval:

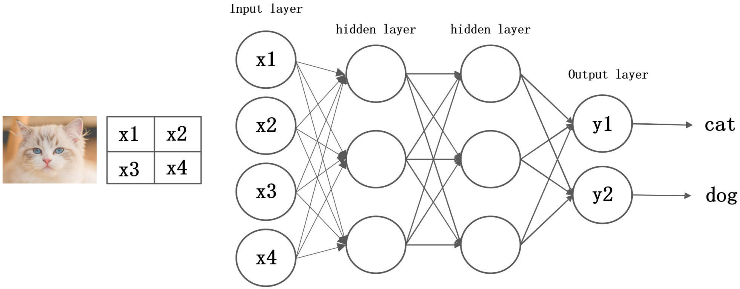
$$x_1 = [0.28, 0.32] \qquad x_2 = [0.18, 0.22] \qquad x_3 = [0.68, 0.72] \qquad x_4 = [0.48, 0.52]$$

- These intervals are then propagated through the neural network.
- Since all calculations are performed on intervals, the resulting outputs y_1 and y_2 will also be intervals.



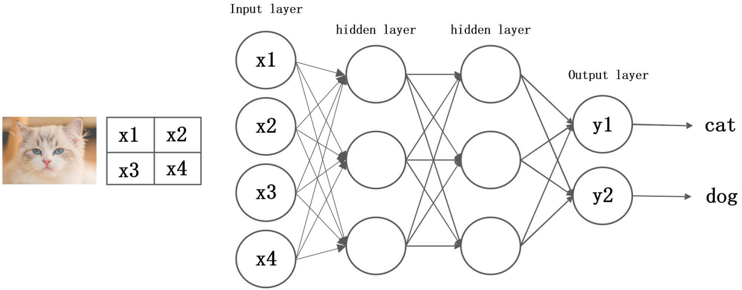
Neural Network Verification Workflow

- If the neural network is robust to perturbations within the ±0.02 range for this cat image, then the lower bound of the y_1 interval should still be greater than the upper bound of the y_2 interval.
- This guarantees that the neural network will consistently classify the image as a cat across all possible ±0.02 range perturbations.



Neural Network Verification Workflow

- The above illustrates the general workflow of neural network verification.
- A verifier is essentially a tool that can perform neural network verification workflow.
- It's worth to note that this neural network verification example is intentionally simplified for clarity and does not capture all technical nuances of formal verification methods.



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Terminology and Evaluation Metrics Explanation

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- **Instance**—An instance consists of a neural network, an input image, a property specification (e.g. Robustness), and a timeout.
- For example, one instance might consist of a classifier for dog and cat, a cat image, a given perturbation range ±0.02, the requirement that output y_1 ("cat") should be greater than y_2 ("dog"), and a specific timeout.

- **Benchmark**—A benchmark consists of a set of related instances.
- Therefore, the benchmark provides predefined networks, images, property specifications, and timeouts. To evaluate a verifier, we simply run it on the benchmark without defining these elements ourselves.

Terminology and Evaluation Metrics Explanation

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- **Verified**—The verifier successfully proves that for all pixel perturbations within the ±0.02 range, the neural network consistently outputs "cat".
- This is the ideal outcome. It means the verifier can deterministically confirm that the neural network operates safely under the specified threat model (such as pixel perturbations within the ±0.02).
- Verifiers such as α-β-CROWN will give the number of successfully verified instances.

Terminology and Evaluation Metrics Explanation

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- **Falsified**—The verifier successfully finds a counterexample—such as a specific set of pixel values (e.g., x_1=0.31, x_2=0.19, . . .)—that causes the neural network to misclassify the "cat" as a "dog".
- This is also a valuable result. It shows that the verifier acts like a detective, uncovering concrete evidence that can "fool" the neural network. This demonstrates the tool's strong attack discovery capabilities.
- Verifiers such as α-β-CROWN will give the number of successfully falsified instances.

Terminology and Evaluation Metrics Explanation

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- **Incorrect Result**—The verifier returns an erroneous conclusion. For example, it may report the network as “safe” (Verified) when a counterexample actually exists (which should have been Falsified).
- This is a negative metric. It indicates potential flaws or inaccuracies in the verifier itself. The fewer incorrect results, the higher the reliability of the verifier.